Improving Early Detection of Cardiac Events with Deep Learning

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Abstract

This study investigates the application of deep learning models to classify heartbeat irregularities using ECG data from the MIT BIH arrhythmia database. The goal is to develop an automated system capable of detecting and diagnosing arrhythmias, including stroke and cardiac arrest indicators, at early stages. Three model architectures were implemented: a one dimensional convolutional neural network, a bi-directional long short-term memory model and a hybrid CNN-BiLSTM model. These models were trained on labeled ECG segments and evaluated for accuracy, precision recall, and F1 score. The proposed system aims to reduce diagnostic latency and variability while offering a scale solution for early detection of critical cardiovascular events.

Introduction

Cardiovascular diseases remain one of the leading causes of death globally, with sudden cardiac arrest and strokes often occurring without prior symptoms. Early detection of abnormal heart rhythms, or arrhythmias, is crucial in reducing mortality and improving patient outcomes. Electrocardiogram (ECG) monitoring is a widely used diagnostic tool in clinical practice; however, accurate interpretation typically requires expert cardiologists and is prone to inter-observer variability. The motivation behind this study is to bridge the gap between ECG data and its effective interpretation by leveraging state-of-the-art deep learning techniques. This study aims to automate the classification of heartbeat irregularities using deep learning models trained on the MIT-BIH Arrhythmia Database.

Literature Review

Numerous studies have applied machine learning to ECG signal classification, with increasing attention to deep learning in recent years. Kiranyaz et al. (2015) demonstrated the feasibility of one-dimensional convolutional neural networks for patient specific arrhythmia classification. Rajpurkar et al. (2017) introduced a deep CNN that achieved cardiologist level performance on an arrhythmia detection using a large scale ECG data set. More recently hybrid models containing CNNs and RNNs, such as those proposed by Yildirim (2018), have shown improved performance and learning both spatial and temporal features of ECG signals. However, many of these models focus on binary classification tasks. This study differentiates itself by using a publicly available dataset and addressing a more granular multi class classification problem that includes both common rare arrhythmias to accurately classify samples of the population, including edge cases. Furthermore, it explores the comparative performance of pure and hybrid deep learning architectures under a consistent training and evaluation pipeline.

Methods

The dataset includes a random sampling of patients, taken from a mixed population with a 3:2 ratio of inpatients to outpatients. The dataset also includes edge-case arrhythmias that are medically significant as well as historical benchmark records.

The beat types used for classification are as follows: Normal Beat (N), Left bundle branch block beat (L), Right bundle branch block beat (R), Atrial premature beat (A), Premature ventricular contraction (V), Paced beat (/), Fusion of ventricular and normal (F), Ventricular escape beat (E), Nodal junctional escape beat (j), and All Others.

Three deep learning models were developed to balance feature extraction and sequence modeling. Training of all models was conducted with TensorFlow and evaluated using multi-class performance metrics. All models used Softmax Dense layers for multi-class classification.

- 1D CNN: This model consists of convolutional layers ranging from 32 to 128 notes, regular activations, batch normalization, and dropout regularization. It is designed to extract spatial features from the ECG signal, like waveform shape and amplitude transitions, which are indicative of different arrhythmia types.
- Bidirectional LSTM: A recurrent neural network architecture used to model temporal dependencies in ECG sequences. The model employs Relu activations, and dropout layers to reduce overfitting. It was designed to capture temporal dependencies and contextual relationships between successive heartbeat segments, which are critical for detecting sequence-level anomalies in rhythm.

- Hybrid CNN-BiLSTM: Combines layers for feature extraction with BiLSTM layers for sequential modeling. This architecture was designed to exploit both spatial patterns and long-term dependences in the ECG waveform.

Results

Model	Accuracy	Precision	Recall	F1-Score
CNN	0.9820			
Top 5 Most Common Classes				
N		0.99	1.00	0.99
L		1.00	0.99	0.99
R		1.00	1.00	1.00
V		0.96	0.98	0.97
/		1.00	0.99	0.99

Model	Accuracy	Precision	Recall	F1-Score
BiLSTM	0.8853			
Top 5 Most Common Classes				
N		0.91	0.97	0.94
L		0.91	0.93	0.92
R		0.83	0.80	0.82
V		0.72	0.78	0.75
/		0.85	0.97	0.91

Model	Accuracy	Precision	Recall	F1-Score
Hybrid	0.9659			
Top 5 Most Common Classes				
N		0.97	1.00	0.98
L		0.99	0.99	0.99
R		0.98	0.99	0.99
V		0.94	0.94	0.94
/		0.98	0.99	0.98

This study proposes three deep learning models: Convolutional Neutral Network, Bidirectional Long Short-Term Memory, and a Hybrid CNN-LSTM model for detecting early cardiac events from ECG signals. Among the models proposed, the CNN achieved the highest test accuracy of 98.2%, demonstrating strong generalization and high precision across the most common arrhythmia classes. The Hybrid model followed with a test accuracy of 96.59%, performing well on major classes but exhibiting reduced precision on low-frequency arrhythmias. Its ROC curve confirms superior discrimination capacity across multiple classes. The BiLSTM model, while effective in capturing temporal dependencies, had the lowest test accuracy oat 88.53%, and struggled especially with minority class detection, as shown by several classes receiving low precision and recall scores. All models performed best on the most common arrhythmias like Normal and Left Bundle Branch, while performance dropped on rare or ambiguous cases. Rare cardiac event detection remains a challenge due to data imbalance. Despite this limitation, the CNN model's speed and accuracy position it as a strong candidate for real-time deployment in clinical settings. Overall, the CNN's superior performance suggests its suitability for large-scale ECG analysis and real-time arrhythmia detection.

Conclusion

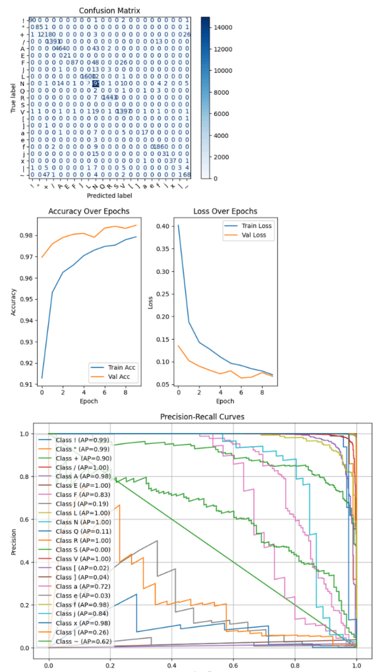
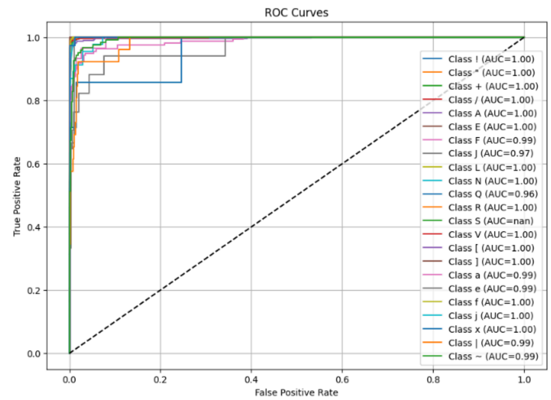
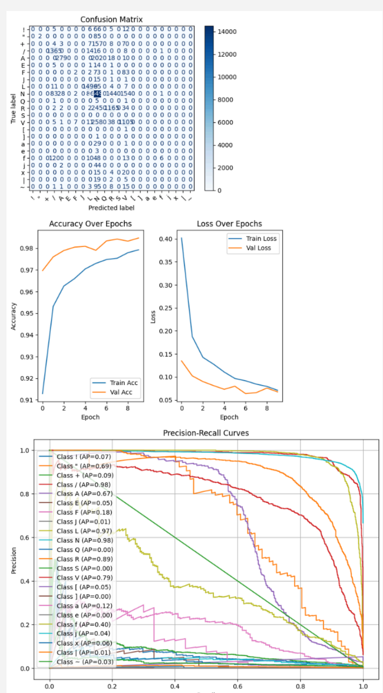
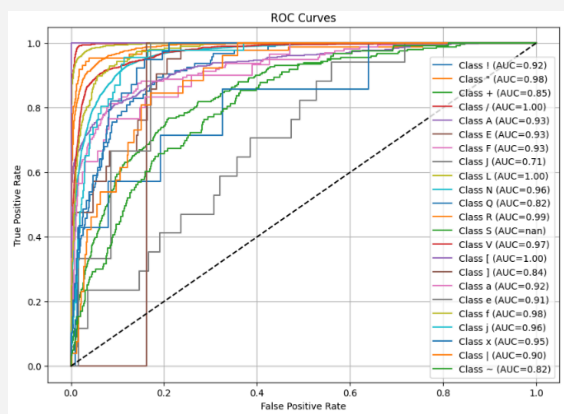
This research aimed to identify an optimal deep learning architecture for automated arrhythmia detection using ECG data. The results indicate that convolutional neural networks are the most effective model among those tested, excelling in both accuracy and class-specific performance. CNN's ability to extract local patterns from ECG signals gives it a significant advantage in recognizing the waveform characteristics associated with various arrhythmias. The authors of this study recommend piloting the CNN model in a controlled hospital environment, focusing on real-time integration and continuous monitoring systems.

While the Bidirectional LSTM model was hypothesized to leverage temporal sequence data, it fell short in both convergence speed and minority class recognition. The hybrid CNN-LSTM model provided a balanced alternative but did not surpass the CNN’s overall performance. These findings suggest that CNN-based architectures should be prioritized for ECG classification tasks, particularly in clinical decision-support systems where high precision and real-time analysis are critical.

In a business context, this study can result in faster diagnoses in potentially at-risk patients. Faster diagnoses can translate to lower costs, higher throughput, and improved patient outcomes. Deploying this model strategically across both inpatient and outpatient settings can transform cardiac care workflows and mitigate operational bottlenecks like cardiologists on-call. Furthermore, integrating this system with existing hospital EMR systems can streamline triage and reduce administrative workload. Healthcare providers may also benefit from reduced malpractice risk by supplementing physician decisions with a consistent, evidence-based diagnostic tool.

For future research, the authors of this study recommend investing in data augmentation to improve recall for edge cases and rare cardiac conditions. Additionally, partnerships with device vendors or wearable technology (such as the Apple Watch) would facilitate deployment across larger subsets of the population. Regulatory approval would be a hard requirement for commercial rollout. Lastly, extending this approach to multi-lead ECG data and testing in geographically diverse populations would further validate its ability to generalize.

Appendix

<table>
<tr><th>Model</th><th>Confusion Matrix, Accuracy/Loss Over Epochs Graph, Precision-Recall Curves</th><th>ROC Curves</th></tr>
<tr><td>CNN</td><td></td><td></td></tr>
<tr><td>Bi-LSTM</td><td></td><td></td></tr>
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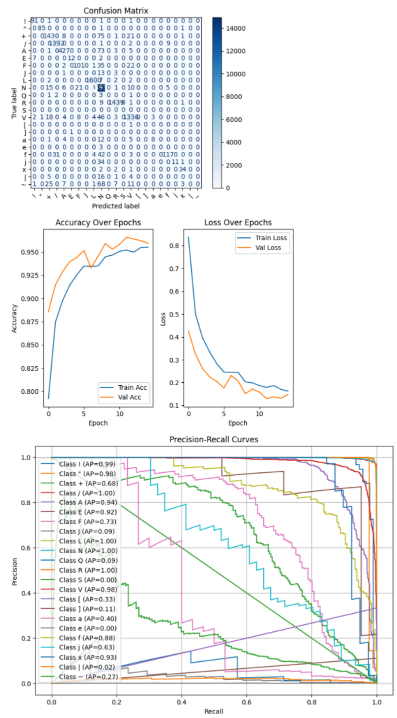
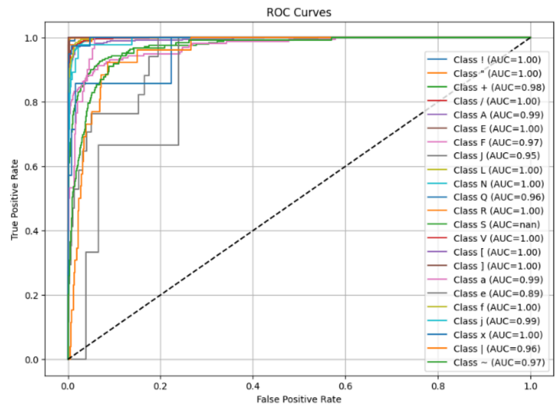
Hybrid	 Confusion Matrix True label Predicted label Accuracy Over Epochs Accuracy Epoch Train Acc Val Acc Loss Over Epochs Loss Epoch Train Loss Val Loss Precision-Recall Curves Precision Recall Class ! (AP=0.99) Class " (AP=0.98) Class + (AP=0.68) Class / (AP=1.00) Class A (AP=0.94) Class E (AP=0.92) Class F (AP=0.73) Class J (AP=0.09) Class L (AP=1.00) Class N (AP=1.00) Class Q (AP=0.09) Class R (AP=1.00) Class S (AP=0.00) Class V (AP=0.98) Class [(AP=0.33) Class] (AP=0.11) Class a (AP=0.40) Class e (AP=0.00) Class f (AP=0.88) Class j (AP=0.63) Class x (AP=0.93) Class | (AP=0.02) Class ~ (AP=0.27)	 ROC Curves True Positive Rate False Positive Rate Class ! (AUC=1.00) Class " (AUC=1.00) Class + (AUC=0.98) Class / (AUC=1.00) Class A (AUC=0.99) Class E (AUC=1.00) Class F (AUC=0.97) Class J (AUC=0.95) Class L (AUC=1.00) Class N (AUC=1.00) Class Q (AUC=0.96) Class R (AUC=1.00) Class S (AUC=nan) Class V (AUC=1.00) Class [(AUC=1.00) Class] (AUC=1.00) Class a (AUC=0.99) Class e (AUC=0.89) Class f (AUC=1.00) Class j (AUC=0.99) Class x (AUC=1.00) Class | (AUC=0.96) Class ~ (AUC=0.97)

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